

Xcelodose 600S- ME-5 Microbalance Failure Investigation

Mechanical failure pathway, observed handling practices, and corrective-action direction

RCA

AUG–SEP 2026



The failure population separates into two distinct patterns

The mechanical pathway is concentrated in the units with visible deformation and corner-load failure.

4

microbalances reviewed

Unavailable between 2023–2026

2

have visible mechanical abnormalities and out-of-service

■ + ■

2

current direct-load observations

17-AUG + 20-AUG

1

most probable mechanical root cause

Handling control during cleaning

Takeaway

Repeated direct force on the fragile weigh pan is the strongest supported mechanical pathway; alignment and powder conditions remain contributing pathways that should be controlled and monitored.

Four failed / unavailable units

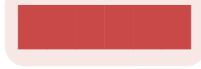
Mechanical Failure RCA basis upon historical evidence and observation.
Software Failure will be alleviated in the next phase of the Replacement and Reliability Plan.

30-JUL-2026 MECHANICAL Corner-load + linearity could not be adjusted Pan tilted left; multiple scuffs	24-AUG-2023 MECHANICAL Failed corner load; could not be repaired TW#151494 05-SEP-2023 Front springs visibly asymmetric	21-JUL-2026 INTERFACE / ELECTRONIC Calibration software connection failure No visible tilt, scuffs, or external damage
---	--	--

RCA boundary Mechanical root-cause analysis: + , and remain separate failure modes requiring their own follow-up.

Mechanical abnormalities are visible on both corner-load failures

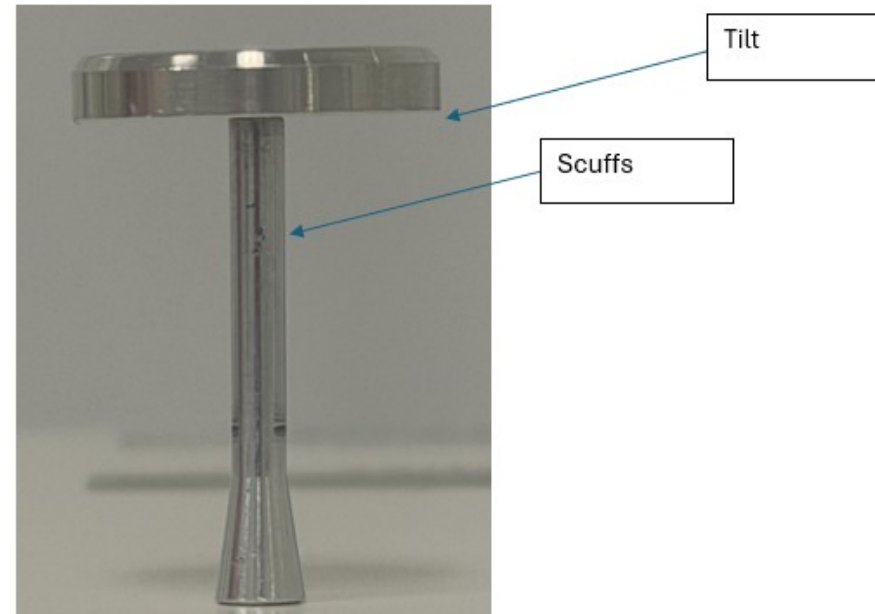
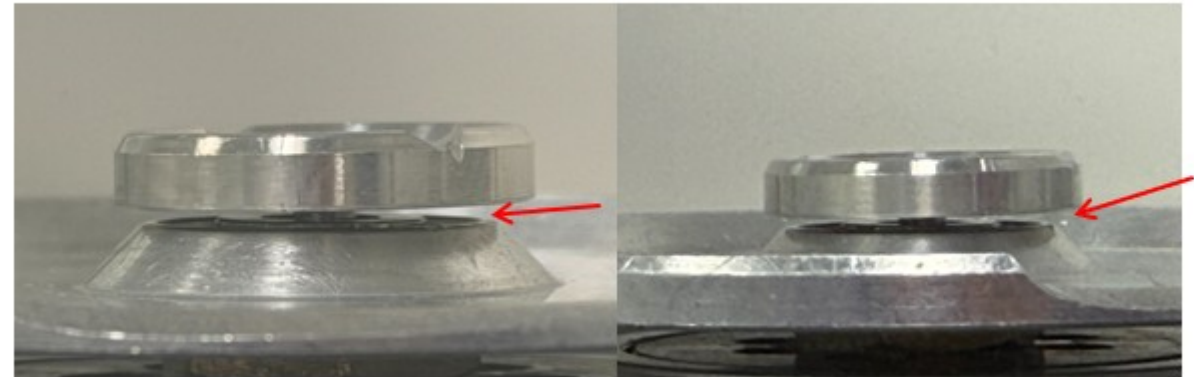
Pan tilted towards left on #0 [REDACTED]



Tilted weigh pan + scuff marks

Failed manufacturer specification; corner load and linearity could not be adjusted.

- Visible geometry is consistent with deformation / misalignment.
- The exact time and loading event that produced the deformation is not directly observed.



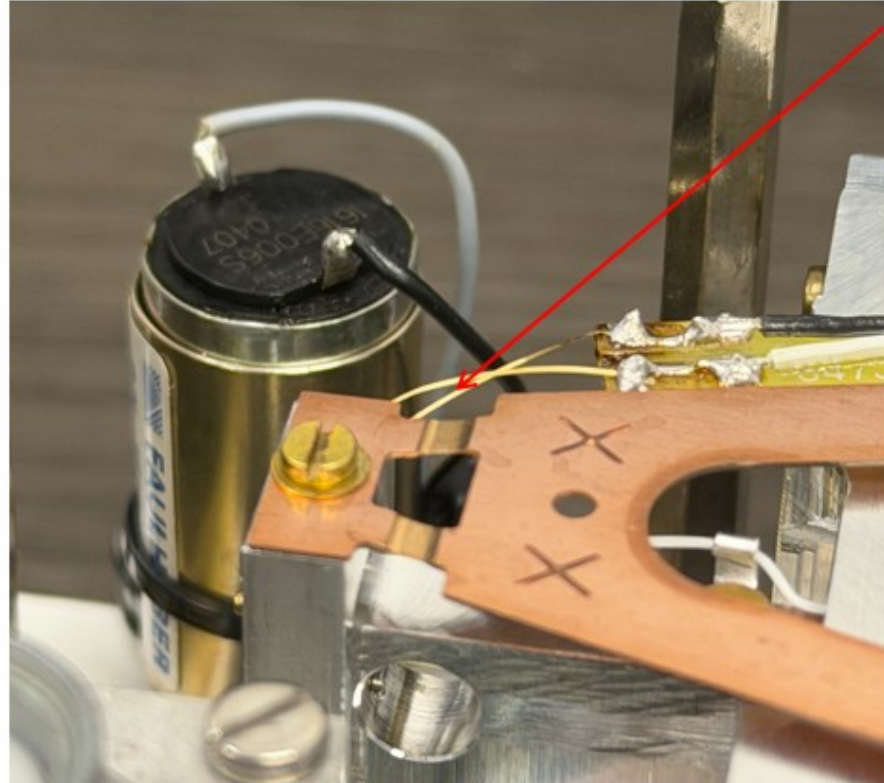
Mechanical abnormalities are visible on both corner-load failures



Asymmetric front springs

Failed corner-load test; unit could not be repaired.

- Visible asymmetry is consistent with spring deformation / mechanical misalignment.
- This unit provides historical evidence of the same corner-load failure family.
- This unit was investigated in TW#151494, direct loading was observed and determined as deformations cause; Pan bend observed similarly to # [REDACTED]



Top front springs are aligned differently, left slopes downward and right side slopes upward.

Triangulate hardware evidence, observed practice, and controlled requirements

No single observation is treated as sufficient on its own, the mechanical cause was built by comparing multiple evidence streams.

FAILED-UNIT INSPECTION

Tilt, scuffs, spring
geometry

MANUFACTURING GEMBA

Operation, cleaning,
handling

SOP / PRACTICE COMPARISON

What is required vs
observed

PRODUCT DEV COMPARISON

Different setup +
cleaning practice

HISTORY + SERVICE

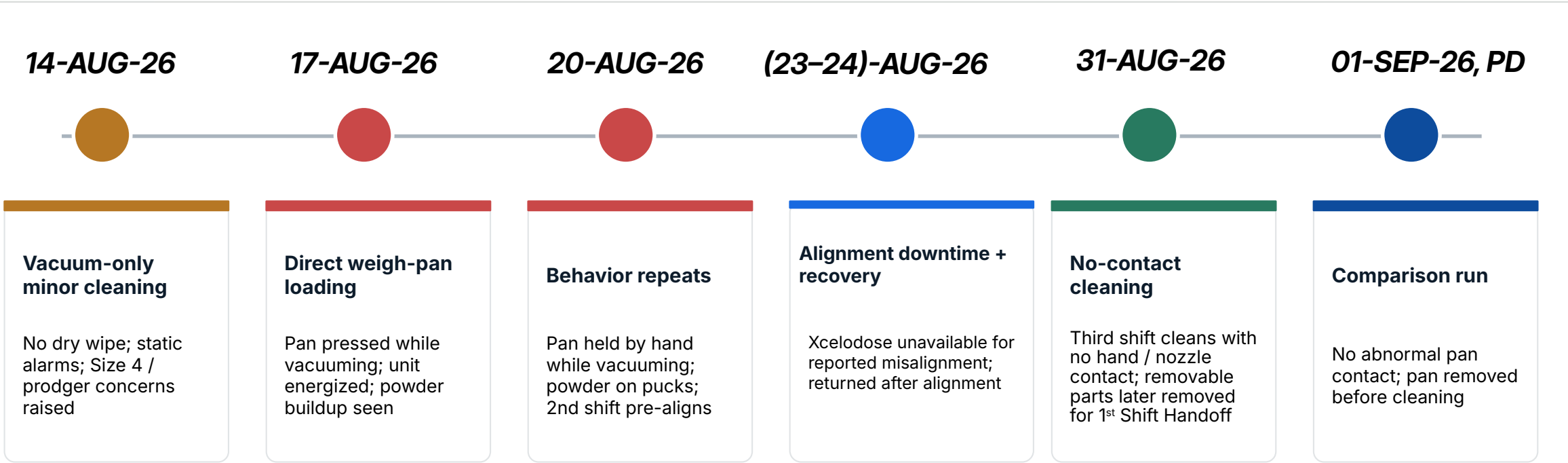
2023 event,
Technician findings

6M + 5-WHY SYNTHESIS

Rank pathways +
system cause

Evidence Flow | Historical Investigation → Observations and SOP Derivations → PD Observations and Comparisons

The strongest evidence emerged across repeated cleaning and alignment observations



Pattern across time The critical handling condition was observed more than once, while later comparison runs demonstrated that cleaning can be performed without direct weigh-pan loading.

Direct weigh-pan force creates a credible load path into the weighing mechanism

This is the only mechanical pathway supported by repeated direct observation plus a relevant historical event.

DIRECT OBSERVATION

**17-
AUG-26**

Operator applied downward force directly to the weigh pan while vacuuming. The system was energized.

Repeated direct-load behavior observed.

DIRECT OBSERVATION

**20-
AUG-26**

The weigh pan was again held by hand during vacuum cleaning.

Confirms the handling behavior recurred.

SERVICE HISTORY

HISTORICAL

TW#151494 documented a mechanical failure associated with pressing the pan during cleaning.

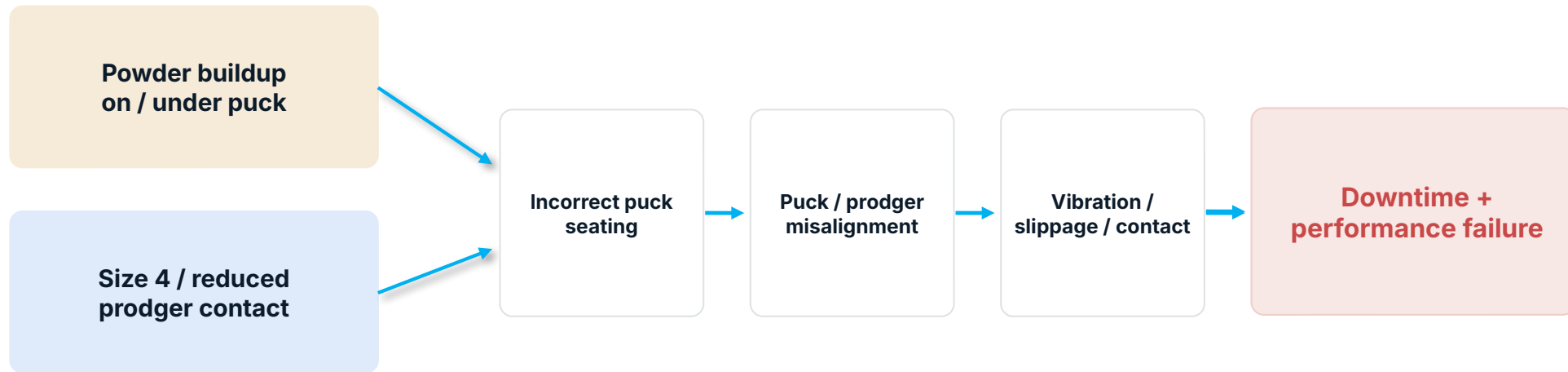
Relevant historical event supports the same load pathway.

What this proves: the loading behavior exists. What still needs manufacturer confirmation: allowable load / damage threshold.

Other:

Powder and alignment conditions may amplify vibration and cause **Downtime**

This pathway is supported by observed buildup and operator / SME input, but it is not as direct as the pan-loading evidence.



EVIDENCE

Observed: powder stuck on pucks (17 / 20 AUG) • Reported: Size 4 slippage + incomplete prodger contact
Observed: pre-alignment performed by 2nd shift • 23-AUG downtime reported as alignment-related

The same cleaning task can be performed without direct weigh-pan loading

MANUFACTURING 1 st / 2 nd Shift — 17 / 20 AUG		THIRD SHIFT — 31 AUG		PRODUCT DEVELOPMENT	
WEIGH-PAN CONTACT	Direct hand restraint / downward force	WEIGH-PAN CONTACT	No hand / vacuum-nozzle contact	WEIGH-PAN CONTACT	No abnormal contact observed
EQUIPMENT STATE	Energized during observed cleaning, Weigh Pan not removed	REMOVABLE PARTS	Removed, wiped, cleaned later	CLEANING METHOD	Weigh pan removed before cleaning
PUCK CONDITION	Powder buildup observed	ALARMS	None observed during cleaning	SME INTERPRETATION	Operational failure most likely when pan is contacted / loaded
RESULTING EVIDENCE	Credible external load path	VALUE OF COMPARISON	Shows no-contact method is feasible	VALUE OF COMPARISON	Reduces opportunity for direct loading

CONTROLLED LEARNING → Direct pan contact is not required to perform the cleaning activity.

Three observed gaps matter because each changes the mechanical risk during cle

01

CONTROL

Dry wipe + vacuum

OBSERVED

Vacuum-only cleaning, weigh pan not removed observed on 14-AUG

RISK LINK

Leaves a greater chance of powder remaining on / around removable parts

02

CONTROL

Protect fragile weigh pan from contact

OBSERVED

Pan held / pressed instead of removal during cleaning on 17-AUG and 20-AUG

RISK LINK

Creates a direct external mechanical load path

03

CONTROL

Clean in required de-energized state

OBSERVED

Observed vacuum cleaning while energized (6.13.1 USTP-608)

RISK LINK

Adds unnecessary interaction with the live mechanism during handling

Positive practice to formalize

2nd-shift pre-alignment and no-contact cleaning technique provide practical controls worth standardizing

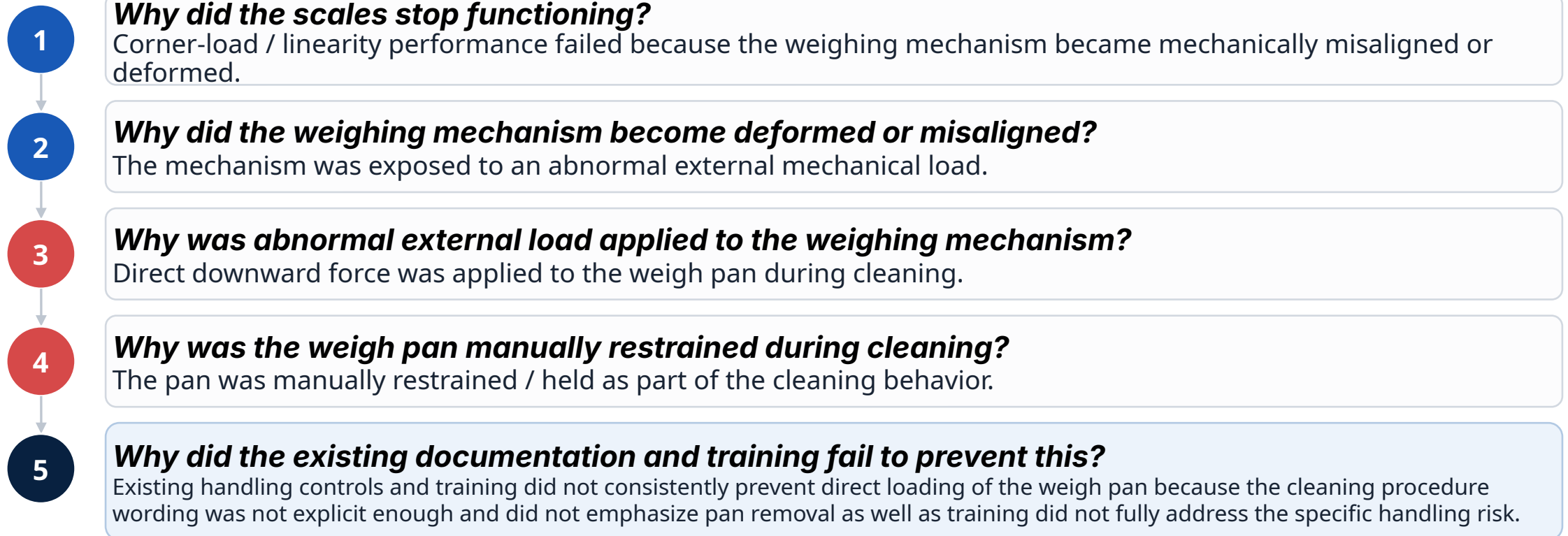
The hypotheses do not carry equal weight



<i>HYPOTHESIS</i>	<i>STATUS</i>	<i>WHY</i>
Direct external load on weigh pan	Evidence-supported	Repeated current observations + historical event
Puck / prodger misalignment	Supported	Alignment downtime, pre-alignment, SME / operator input
Powder buildup → improper seating	Supported	Caked / stuck powder observed on pucks
Operational vibration / contact	Possible	Size 4 slippage + scuff evidence; causal link not fully proven
Vacuum-related loading / static	Possible	Static alarms temporally associated with one cleaning event

Use this ranking to focus CAPA on the supported load path while continuing to trend the secondary contributors.

The 5-Why moves from the failed balance to the control that allowed repeat loading



The system-level failure is not “an operator mistake”; it is an ineffective control that allowed the same fragile-component loading behavior to recur.

Ineffective control of weigh-pan handling during cleaning

Repeated direct loading of the fragile pan provides the clearest supported path to mechanical deformation / misalignment and corner-load failure.

Most Probable Root Cause

Direct manual force was repeatedly applied to the weigh pan during cleaning; The existing handling controls and training did not consistently prevent recurrence.

Mechanical consequence

External load → deformation / misalignment → corner-load and/or linearity failure

Why this outranks the alternatives

- REPEATED** Direct load observed 17-AUG and 20-AUG
- HISTORICAL** 2023 event links pan pressing with mechanical failure
- PHYSICAL** [redacted] and [redacted] show mechanical abnormalities
- COMPARISON** 3rd shift and PD demonstrate no-contact cleaning is feasible

Secondary contributors remain relevant: puck/prodger alignment, powder buildup, Size 4 slippage/vibration, and vacuum/static behavior should continue to be controlled.

Observations and Corrective actions

OBSERVED / DOCUMENTED

- Physical condition of failed balances
- 17 + 20 AUG direct pan loading
- Powder buildup on pucks
- 31 AUG no-contact cleaning
- PD pan-removal practice follows SOP and Weigh Pan removal
- SOP-to-practice differences

SUPPORTED INFERENCE

- Direct load transfers into weighing mechanism
- Powder can interfere with correct seating
- Misalignment can increase contact / vibration
- No-contact cleaning lowers loading opportunity

CONCLUSION / CORRECTIVE ACTION

- Most probable root cause: Training outdated and control gap leading to direct pan force
- Leads only should remove and install the microbalance
- Reinforcement and Retraining Initiative
- Pan removal reinforced in SOP, update language. Bold section 6.3.2.11.2 "Remove Weighing pan and spill tray..."; Add Pan removal to section 6.13.2 Minor Cleaning. All for USTP-608
- MSAT Engineers get Method Development/
Super-User role and Trained Leads given Operator/Super-User

Primary internal sources: Xcelodose investigation, current RCA • Historical findings • 14–31 AUG Manufacturing observations
• Product Development comparison • SOP review vs Observations

Control the load path first, then standardize alignment and corrective action



CONTAIN NOW

Reduce immediate exposure

- No direct force on weigh pan
- Adjust SOP and required equipment state to include pan removal during cleaning
- Keep pan isolated from vacuum / direct force

STANDARDIZE

Remove method variation

- Define cleaning procedure and removable-part cleaning
- Clarify pre-alignment ownership
- Assign MSAT to Method Dev/Super user and remove any Operator from roles with parameter adjustment

TRAIN

Make correct handling repeatable

- Demonstrate proper cleaning and correct pan removal
- Qualify operator / super-user alignment role
- Training and SOP updates for Pan removal and no contact cleaning

ACT

Close remaining technical gaps

- Update SOP Terminology and Cleaning sections
- Inspect Operations after training
- Installation and Removal of balance by leads only.

Implementation principle → engineer out the contact opportunity where practical; do not rely on memory alone.

The mechanical pathway is actionable now

Formalize Controls, assign ownership and verify effectiveness

1

Approve the cleaning / handling control direction

No direct weigh-pan force; define removal / isolation and equipment-state requirements.

2

Assign alignment ownership + training

Decide who can perform pre-alignment, removal and installation;
As well as how competency is documented.

3

Implement physical damage-prevention controls

Lead operator routine inspections. Leads verify retraining effectiveness and document repeat pan contact, misalignment, or damage.

4

Track recurrence and reliability performance

Document scale failures, alignment issues, damage events, and repeat maintenance calls and confirm whether the new controls are reducing failures over time.

CURRENT CONCLUSION

Most probable root cause: ineffective control of direct weigh-pan handling during cleaning.